THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a 100% stacked bar of a household's monthly water use, appliances take 33% of 1000 units. The units used by the appliances are: 1

(a) 329 **(b)** 332 **(c)** 331 **(d)** 330

2. The average speed over a return trip on the same route at two different speeds is: 1

(a) total distance divided by total time **(b)** the mean of the two speeds
(c) the difference of the speeds **(d)** the larger speed

3. The average of 8 numbers is 33. The average of all 32 numbers (including 24 more numbers) is 60. The average of the 24 additional numbers is: 1

(a) 69 **(b)** 70 **(c)** 68 **(d)** 71

4. The average of 15 numbers is 37. If the number 65 is removed, the average of the remaining 14 numbers is: 1

(a) 37 **(b)** 34 **(c)** 36 **(d)** 35

5. The average marks of 38 students is 76 and of another 44 students is 35. The average of all 82 students is: 1

(a) 55 **(b)** 56 **(c)** 54 **(d)** 53

6. The average of 12 numbers is 46. When a number 59 is included, the average of all 13 numbers becomes: 1

(a) 49 **(b)** 46 **(c)** 48 **(d)** 47

7. Marks of 66, 78 and 63 are given weights 2, 4 and 5 respectively. The weighted average is: 1

(a) 70 **(b)** 68 **(c)** 69 **(d)** 71

8. 180 mL of a 25% sugar solution is mixed with 90 mL of a 28% sugar solution. The concentration of the mixture is: 1

(a) 26% **(b)** 27% **(c)** 28% **(d)** 25%

9. In a 100% stacked bar of a household's monthly water use, appliances take 38% of 800 units. The units used by the appliances are: 1

(a) 306 **(b)** 305 **(c)** 304 **(d)** 303

10. If a new value larger than the current mean is added to a data set, the mean: 1

(a) increases **(b)** decreases **(c)** stays the same **(d)** becomes zero

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A 100% stacked bar of a school's ₹1400 budget shows 27% on books and 14% on sports. Find the amount on books, on sports and on the remaining items. 2

12. The average marks of 46 students in a test is 77 and that of 30 students is 39. Find the average marks of all 76 students together. 2

13. A final score combines Maths (score 82, weight 5), Science (score 45, weight 3), English (score 73, weight 5). Find the weighted average score. 2

14. A 100% stacked bar of a school's ₹900 budget shows 39% on books and 11% on sports. Find the amount on books, on sports and on the remaining items. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 130 mL of a 36% sugar solution is mixed with 260 mL of a 15% sugar solution. Find the percentage concentration of the mixture. 3

16. The mean of 19 observations is 40. One observation 94 is removed. Find the mean of the remaining observations. 3

17. The average score of 30 students in a test is 53. When 32 more students take the test, the average score of all 62 students becomes 37. Find the average score of the 32 additional students. 3

OR

180 mL of a 27% sugar solution is mixed with 270 mL of a 32% sugar solution. Find the percentage concentration of the mixture. 3

18. The mean of 11 observations is 59. One observation 49 is removed. Find the mean of the remaining observations. 3

OR

The average score of 27 students in a test is 35. When 18 more students take the test, the average score of all 45 students becomes 51. Find the average score of the 18 additional students. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 28, 44 and 21 students with average marks 48, 45 and 41 respectively. Find the average marks of all the students. 5

OR

A family spent ₹3600 in January and ₹3900 in February. Food took 33% of the January total and 32% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

20. Three sections have 28, 24 and 20 students with average marks 67, 48 and 61 respectively. Find the average marks of all the students. 5

OR

A family spent ₹2000 in January and ₹3000 in February. Food took 54% of the January total and 21% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks